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United States Patent	12402304
Kind Code	B2
Date of Patent	August 26, 2025
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Semiconductor structure and manufacturing method thereof

Abstract

Embodiments of the present disclosure relate to a semiconductor structure and a manufacturing method thereof. The method of manufacturing a semiconductor structure includes: providing a substrate; etching the substrate to form a plurality of body structures arranged at intervals in a first direction, wherein a space between adjacent ones of the body structures is filled with a first isolation layer, each of the body structures includes a body layer and a plurality of body pillars that are discrete and are located on the body layer, the plurality of body pillars are arranged at intervals along a second direction, and the first direction is different from the second direction; etching a part of the body layer between adjacent ones of the body pillars, to form a plurality of openings in the body layer; and siliconizing the body layer through the plurality of openings.

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Appl. No.:	18/153459
Filed:	January 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230354584 A1	Nov. 02, 2023

Foreign Application Priority Data

CN	202210467904.X	Apr. 29, 2022
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Publication Classification

Int. Cl.: H10B12/00 (20230101); H10D30/67 (20250101)

U.S. Cl.:

CPC H10B12/482 (20230201); H10B12/05 (20230201); H10B12/488 (20230201);
H10D30/6735 (20250101);

Field of Classification Search

CPC: H10B (12/482)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority of Chinese Patent Application No. 202210467904.X, submitted to the Chinese Intellectual Property Office on Apr. 29, 2022, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) Embodiments of the present disclosure relate to the technical field of integrated circuit design and manufacturing, and in particular, to a semiconductor structure and a manufacturing method thereof.

BACKGROUND

(3) With the higher level of integration and performance of memory, a vertical gate-all-around (VGAA) transistor can effectively reduce the size of the memory. It is important to ensure the reliable performance of the memory with VGAA transistors.

SUMMARY

(4) A first aspect of the embodiments of the present disclosure provides a method of manufacturing a semiconductor structure, including: providing a substrate; etching the substrate to form a plurality of body structures arranged at intervals in a first direction, wherein a space between adjacent ones of the body structures is filled with a first isolation layer, each of the body structures includes a body layer and a plurality of body pillars that are discrete and are located on the body layer, the plurality of body pillars are arranged at intervals along a second direction, and the first direction is different from the second direction; etching a part of the body layer between adjacent ones of the body pillars, to form a plurality of openings in the body layer; and siliconizing the body layer through the plurality of openings to form a metal silicide that is continuous along the second direction, so as to form a plurality of bit lines, each of the bit lines being in a corresponding body layer.

(5) The embodiments of the present disclosure further provide a semiconductor structure, including: a substrate; a plurality of body structures arranged at intervals along a first direction, wherein a space between adjacent ones of the body structures is filled with a first isolation layer, each of the body structures includes a body layer and a plurality of body pillars that are discrete and are located on the body layer, the plurality of body pillars are arranged at intervals along a second direction, and a part of the body layer between adjacent ones of the body pillars is provided with openings; and a plurality of bit lines, each of the bit lines including a metal silicide that is located in a corresponding body layer and extends along the second direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To better describe and illustrate the embodiments and/or examples of the present disclosure, reference may be made to one or more accompanying drawings. Additional details or examples for describing the drawings should not be considered as limitations on the scope of any one of the present disclosure, the currently described embodiment and/or example, and the optimal mode of the present disclosure as currently understood.

(2) FIG. 1 is a flowchart of a method of manufacturing a semiconductor structure according to an embodiment of the present disclosure.

(3) FIG. 2 to FIG. 14 are schematic structural diagrams corresponding to a method of manufacturing a semiconductor structure according to an embodiment of the present disclosure, wherein direction ox is a first direction, direction oy is a second direction, and direction oz is a height/thickness direction.

(4) FIG. 15 is a flowchart of a method of manufacturing a semiconductor structure according to another embodiment of the present disclosure.

(5) FIG. 16 to FIG. 23 are schematic structural diagrams corresponding to a method of manufacturing a semiconductor structure according to another embodiment of the present disclosure, wherein direction ox is a first direction, direction oy is a second direction, and direction oz is a height/thickness direction.

REFERENCE NUMERALS

(6) **100**: substrate; **101**: first patterned mask layer; **102**: first trench; **201**: second patterned mask layer; **202**: second trench; **10**: body structure; **11**: body layer; **11a**: opening; **12**: body pillar; **13**: protective layer; **13a**: gap; **20**: first isolation layer; **30**: bit line; **40**: VGAA transistor; **41**: gate; **411**: gate dielectric layer; **412**: gate conductive layer; **50**: word line; **60**: second isolation layer; **70**: third isolation layer; **80**: fourth isolation layer.

DETAILED DESCRIPTION

(7) To facilitate the understanding of the present disclosure, the present disclosure is described more completely below with reference to the related accompanying drawings. The preferred embodiments of the present disclosure are shown in the accompanying drawings. However, the present disclosure may be embodied in various forms without being limited to the embodiments described herein. On the contrary, these embodiments are provided to make the present disclosure more thorough and comprehensive.

(8) Unless otherwise defined, all technical and scientific terms used in the specification have the same meaning as commonly understood by those skilled in the technical field of the present disclosure. The terms used in the specification of the present disclosure are merely for the purpose of describing specific embodiments, rather than to limit the present disclosure. The term “and/or” used herein includes any and all combinations of one or more of the associated listed items.

(9) It should be noted that, the drawings provided in the embodiments merely illustrate the basic concepts of the present disclosure schematically. Although the drawings only show components related to the present disclosure rather than being drawn according to the quantities, shapes, and sizes of components in actual implementation, patterns, quantities, and proportions of components in actual implementation may be changed randomly, and the component layout may be more complex.

(10) Embodiments of the present disclosure provide a semiconductor structure and a manufacturing method thereof. A continuous metal silicide is formed in a substrate to serve as a buried bit line, to reduce the resistance of the semiconductor structure and improve the performance of the semiconductor structure. The semiconductor structure is applicable to a VGAA transistor, to effectively reduce the size of a memory and improve the level of integration and performance of the memory.

(11) The embodiments of the present disclosure provide a method of manufacturing a semiconductor structure, which is described in detail below with reference to the drawings. FIG. 1 to FIG. 23 are schematic structural diagrams corresponding to a method of manufacturing a semiconductor structure according to an embodiment of the present disclosure. It should be noted that, to conveniently describe and clearly illustrate the method of manufacturing a semiconductor structure, the drawings in this embodiment are all partial schematic structural diagrams of the semiconductor structure.

(12) FIG. 2 to FIG. 6 are three-dimensional schematic structural diagrams; FIG. 7 and FIG. 14 are partial cross-sectional views taken along a first direction ox; and FIG. 8 to FIG. 13 are partial cross-sectional views taken along a second direction oy.

(13) Referring to FIG. 1, in some embodiments of the present disclosure, a method of manufacturing a semiconductor structure is provided, which includes:

(14) Step S10: Provide a substrate **100**.

(15) Step S11: Etch the substrate **100** to form a plurality of body structures **10** arranged at intervals in a first direction ox, wherein a space between adjacent body structures **10** is filled with a first isolation layer **20**, each body structure **10** includes a body layer **11** and a plurality of discrete body

pillars **12** located on the body layer **11**, the plurality of body pillars **12** are arranged at intervals along a second direction oy, and the first direction ox is different from the second direction oy.

(16) Step **S12**: Etch a part of the body layer **11** between adjacent body pillars **12** to form a plurality of openings **11a** in the body layer **11**.

(17) Step **S13**: Siliconize the body layer **11** through the plurality of openings **11a** to form a metal silicide that is continuous along the second direction, so as to form a plurality of bit lines **30**, each bit line **30** is in a corresponding body layer **11**.

(18) Referring to FIG. **1** and FIG. **2**, in some embodiments of the present disclosure, the substrate **100** provided in step **S10** may be formed by a semiconductor base such as silicon wafer. The material of the substrate **100** includes, but is not limited to, monocrystalline silicon, polycrystalline silicon or amorphous silicon; the substrate **100** includes, but is not limited to, a monocrystalline silicon substrate, a polycrystalline silicon substrate, or an amorphous silicon substrate.

(19) Further referring to FIG. **1** to FIG. **4**, in step **S11**, in some embodiments of the present disclosure, a plurality of body structures **10** arranged at intervals along the first direction ox may be formed on the substrate **100** by etching the substrate **100**, and a space between adjacent body structures **10** is filled with the first isolation layer **20**.

(20) Referring to FIG. **2** to FIG. **4**, a first patterned mask layer **101** is formed on the substrate **100**, wherein parts of the first patterned mask layer **101** are arranged at intervals along the first direction ox and extend along the second direction oy. The substrate **100** is etched by using the first patterned mask layer **101** as a mask, to form a plurality of first trenches **102** in the substrate **100**. Then, the first patterned mask layer **101** is removed and a deposition process is performed, to form an isolation material that covers a top surface of the substrate **100** and fills up the first trenches **102**. A chemical mechanical planarization process is performed on the isolation material to expose the top surface of the substrate **100**, such that the first trenches **102** are filled with the first isolation layer **20**. The isolation material is not particularly limited, and may be, for example, silicon oxide or silicon nitride.

(21) Referring to FIG. **5** and FIG. **6**, a second patterned mask layer **201** is formed on a surface jointly formed by the substrate **100** and the first isolation layer **20**. Parts of the second patterned mask layer **201** are arranged at intervals along the second direction oy different from the first direction ox, and extend along the first direction ox. The substrate **100** and the first isolation layer **20** are etched by using the second patterned mask layer **201** to form a plurality of second trenches **202**, wherein an etching depth of the second trench **202** is less than an etching depth of the first trench **102**. The second patterned mask layer **201** is removed, to form a plurality of body structures **10** arranged at intervals in the first direction ox on the substrate **100**. Each body structure **10** includes a body layer **11** and a plurality of discrete body pillars **12** on the body layer **11**. The plurality of body pillars **12** are arranged at intervals along the second direction oy and spaced apart by the plurality of second trenches **202**.

(22) Further referring to FIG. **6**, the region of the body layer **11** is used for forming a bit line **30** subsequently. In the embodiments of the present disclosure, the bit line **30** has low resistance, and the manufactured semiconductor structure has desirable performance. A component device of the body pillar **12** may be a junctionless transistor. The body pillar **12** includes a source, a vertical channel, and a drain that are arranged in sequence. The source, the vertical channel, and the drain are doped with ions of a same type. On one hand, controllability of the gate of the transistor can be ensured, to improve the integration density and electrical performance of the semiconductor structure. On the other hand, impact caused by the growth of the bit line **30** can be minimized, thereby avoiding finally affecting the performance of the semiconductor structure.

(23) Further referring to FIG. **6**, in some embodiments of the present disclosure, the first direction ox intersects with the second direction oy. An angle between the first direction ox and the second direction oy is greater than 0° and less than or equal to 90°. For example, the angle is 1°, 2°, 10°, 30°, 50°, 70° or 90°. Optionally, the first direction ox and the second direction oy are perpendicular

to each other.

(24) Referring to FIG. 3 and FIG. 6 to FIG. 8 again, in some embodiments of the present disclosure, the first trenches **102** each located between adjacent ones of the plurality of body structures **10** extend along the second direction oy and are arranged at intervals along the first direction ox, and can separate the plurality of bit lines **30** from each other to form bit line trenches. The second trenches **202** each located between adjacent ones of the plurality of body pillars **12** extend along the first direction ox and are arranged at intervals along the second direction oy, and can separate the plurality of word lines **50** from each other. The etching depth of the second trench **202** is less than the etching depth of the first trench **102**. For example, the etching depth of the first trench **102** is 2000 Angstrom to 4000 Angstrom, such as 2000 Angstrom, 3000 Angstrom, or 4000 Angstrom. When the second trench **202** is excessively deep, space for forming the bit line **30** subsequently is insufficient, and forming of the openings **11a** in the body layer **11** is also affected, making it difficult to manufacture the bit line **30**. When the second trench **202** is excessively shallow, the height of the body pillar **12** is reduced, which also results in insufficient space for forming the word line **50** and the VGAA transistor. Therefore, the depth of the second trench **202** is, for example, 1000 Angstrom to 2000 Angstrom, such as 1000 Angstrom, 1500 Angstrom, or 2000 Angstrom. In the embodiments of the present disclosure, the depth of the second trench **202** is less than the depth of the first trench **102**, to ensure that the plurality of body layers **11** are arranged at intervals in parallel along the first direction ox, thereby ensuring the plurality of bit lines **30** subsequently formed and each being in the corresponding body layer **11** are independent of each other, thus avoiding unexpected situations such as electric leakage.

(25) Referring to FIG. 7, FIG. 8, and FIG. 10, in step S12, the part of the body layer **11** located between adjacent body pillars **12** is etched, to form a plurality of openings **11a** in the body layer **11**. In some embodiments, the opening **11a** is, for example, an arc-shaped opening. Orthographic projection of the arc-shaped opening **11a** on an upper surface of the substrate **100** partially overlaps with orthographic projections of two body pillars **12**, that are adjacent to the arc-shaped opening **11a**, on the upper surface of the substrate **100**. That is, the plurality of openings **11a** are arranged at intervals along the second direction oy in the body layer **11**. Adjacent openings **11a** are close to but not in contact with each other under the body pillar **12**. A maximum opening distance of the arc-shaped opening **11a** in the body layer **11** is 8.5 nm to 11.5 nm, for example, 8 nm, 9 nm or 10 nm, which not only ensures the stability of the body pillar **12** located above, but also helps the metal silicide to grow from the opening **11a** to both sides sufficiently during subsequent metal silicide processing in the body layer **11** through the openings **11a**, so as to form the metal silicide that is continuous in the second direction oy to serve as the bit line **30**. Certainly, in other embodiments of the present disclosure, the opening **11a** may also be in other shapes, e.g., strip-shaped or polygon-shaped. It should be noted that, in the embodiments of the present disclosure, the opening **11a** is first formed in the part of the body layer **11** between adjacent body pillars **12**, which is adaptable to the case where the spacing between the body pillars **12** is insufficient while the bit line **30** needs to be grown through the spacing between the body pillars **12**. In this way, the bit line **30** is continuous in the second direction oy.

(26) Referring to FIG. 9, FIG. 10, and FIG. 12, in some embodiments of the present disclosure, the bit lines **30** may be formed before the transistors **40** and word lines **50** are formed. In order to project channel sidewalls of the body pillars **12** for subsequently manufacturing the transistors **40**, sidewalls of the body pillars **12** may be projected.

(27) Referring to FIG. 9, a protective layer **13** is formed on the sidewalls of the body pillars **12**, and gaps **13a** exist between parts of the protective layer **13** of adjacent body pillars **12**. When the plurality of body pillars **12** are separated from each other by the plurality of second trenches **202**, a protective layer material may be deposited on surfaces of the plurality of second trenches **202**. The protective layer material covers sidewalls and bottoms of the second trenches **202** as well as top surfaces of the body pillars **12**. Then, a part of the protective layer at the bottoms of the second

trenches **202** and on the top surfaces of the body pillars **12** is removed through, for example, an etching process, so as to form a protective layer **13** on the sidewalls of the body pillars **12**. Gaps **13a** exist between parts of the protective layer **13** of adjacent body pillars **12** to expose the part of the body layer **11** between the adjacent body pillars **12** and the top surfaces of the body pillars **12**. In some embodiments of the present disclosure, the protective layer **13** may be made of, for example, an insulating material such as silicon oxide, silicon nitride, or silicon oxynitride.

(28) Referring to FIG. **10**, an etching process, such as isotropic etching, may be performed on the body layer **11** along the gaps **13a**, to form the plurality of openings **11a** arranged at intervals along the second direction oy in the body layer **11**. The isotropic etching has no obvious directional selectivity, such that the arc-shaped openings **11a** can be formed in the body layer **11**. Adjacent openings **11a** are close to but not in contact with each other under the body pillar **12**, which not only ensures the stability of the body pillar **12** located above, but also facilitates growth of a continuous metal silicide in the body layer **11** through the openings **11a**. In some embodiments, for example, a reactive etching gas, such as Cl.sub.2, may be used in plasma etching to improve the etching speed and etching profile.

(29) In other embodiments of the present disclosure, after the openings **11a** are formed, the openings **11a** may be cleaned by using any one or any combination of SC1 solution, SC2 solution, SC3 solution, hydrofluoric acid or dry CF.sub.4 gas, to improve cleanliness in the openings **11a**, thereby improving the conductivity of the bit line **30** subsequently formed by annealing the conductive material layer deposited on the surfaces of the openings **11a**.

(30) Referring to FIG. **12**, in step S40, in some embodiments, the body layer **11** is siliconized through the plurality of openings **11a** to form a metal silicide that is continuous along the second direction oy, so as to form a plurality of bit lines **30**, each bit line **30** is in a corresponding body layer **11**.

(31) Further referring to FIG. **12**, a conductive material is deposited on the surfaces of the plurality of openings **11a**, and an annealing process is performed to form the bit line **30**. The conductive material is deposited, by using a chemical vapor deposition (CVD) process for example, into the openings **11a** along the spacing between the body pillars **12**, for example, along the gaps **13a**. Then, annealing is performed, such that the conductive material is siliconized by the silicon material of the body layer **11** to form a metal silicide. Afterwards, the unreacted metal silicide may further be removed. The openings **11a** can increase a reaction area between the conductive material and silicon, such that the grown metal silicide is continuous along the second direction oy in the body layer **11** to form the bit line **30**. In some embodiments, the conductive material may be a metal material such as Co, Ni, Ti, W, Cu, or Al.

(32) Referring to FIG. **11** and FIG. **12**, in other embodiments of the present disclosure, to reduce the Schottky contact resistance between the bit line **30** and the substrate **100** as well as the body structure **10**, that is, the Schottky contact resistance at the interface between the metal silicide and the silicon, the following may be performed before step S13: performing an ion implant on the body layer **11** along the plurality of openings **11a** with a preset energy and a preset dosage to form a doped region **11b**, and performing an annealing process; then siliconizing the body layer having the doped region **11b** to form the bit line **30**.

(33) Further referring to FIG. **11**, in some specific embodiments, for example, ion implant may be implemented through ion implantation using P-type ions, such as B ions. Definitely, in other embodiments, N-type ions that have a higher current may be used. Specifically, for example, As ions or P ions may be used. The preset energy for ion implantation may be, for example, 1 keV to 20 keV; and/or the preset dosage is 1e.sup.15 cm.sup.-2 to 1e.sup.18 cm.sup.-2. For example, the preset energy is 1 keV, 1.5 keV, 5 keV, 10 keV or 12 keV; the preset dosage is 1e.sup.15 cm.sup.-2, 5e.sup.15 cm.sup.-2, 1e.sup.16 cm.sup.-2, 1e.sup.17 cm.sup.-2 or 1e.sup.18 cm.sup.-2. In the embodiments of the present disclosure, at least one ion implantation operation within the foregoing range, e.g., one low-energy high-dosage ion implantation operation, is performed on the openings

11a at the bottoms of the gaps between adjacent body pillars **12** and on the tops of the body pillars **12**, such that impurities are accumulated at the interface between the silicide and silicon due to the segregation effect in the process of forming the metal silicide, thereby reducing the Schottky contact resistance and improving the performance of the semiconductor structure. In other embodiments of the present disclosure, the protective layer **13** provides little projection effect for the sidewalls of the body pillars **12**. In addition, to further reduce the impact on the body pillar **12**, the subsequently formed transistor **40** surrounding the body pillar **12** may be a junctionless transistor (e.g., referring to FIG. **13**), such that after the body pillar **12** serving as the vertical channel is doped with a certain dosage of ions, the body pillar **12** itself is highly doped, for example, the dosage is $1\text{e.sup.}18\text{ cm.sup.}^{-3}$ to $1\text{e.sup.}20\text{ cm.sup.}^{-3}$. Therefore, the ion implantation in the embodiments of the present disclosure has little impact on the subsequently manufactured transistor **40**.

(34) Referring to FIG. **13** and FIG. **14**, in some embodiments of the present disclosure, after the plurality of bit lines **30** each being in the corresponding body layer **11** are formed, semiconductor devices may further be manufactured on the substrate **100** and the body structure **10** subsequently.

(35) Referring to FIG. **13** and FIG. **14**, the plurality of body pillars **12** on the body structure **10** are used as active regions to form a plurality of corresponding VGAA transistors **40**. A gate **41** of the VGAA transistor **40** surrounds the body pillar **12**. The plurality of body pillars **12** are arranged at intervals along the second direction oy and are spaced apart from each other by the plurality of second trenches **202**. In the embodiments of the present disclosure, for example, the VGAA transistor **40** may be a junctionless transistor, which can not only improve the performance of the semiconductor structure to ensure the control capability of the gate **41**, but also avoid impact on the VGAA transistor **40** during forming of the bit line **30**. Specifically, for example, the body pillar **12** is formed into a source (not shown in the figure), a vertical channel, and a drain (not shown in the figure) that are sequentially arranged by doping the body pillar **12** with ions of a same type. One of the source of the VGAA transistor or the drain of the VGAA transistor **40** is connected to the bit line **30**, and the other is connected to a capacitor (not shown in the figure). The doping process of the body pillar **12** may be performed simultaneously with, or performed before or after the ion implantation process, which is not particularly limited and may be selected according to actual process requirements.

(36) Further referring to FIG. **13** and FIG. **14**, a conductive material is deposited in the second trenches **202** to manufacture the gates **41** and the word lines **50**. In some embodiments, for example, the protective layer **13** on the sidewalls of the body pillars **12** and the first isolation layer **20** may be removed, and the second trenches **202** are filled with the conductive material. A gate dielectric layer **411** may be formed before the conductive material is formed. For example, the surfaces of the body pillars **12** may be oxidized to form the gate dielectric layer **411**, where the gate dielectric layer **411** surrounds the body pillars **12**. Then, the conductive material is deposited, and the conductive material is planarized and etched back to form the word lines **50**. The conductive material around the gate dielectric layer **411** forms a gate conductive layer **412**, where the gate dielectric layer **411** and the gate conductive layer **412** constitute the gates **41** of the transistors **40**. Each word line **50** connects the gates **41** of the VGAA transistors **40** and extends continuously along the first direction ox; the word lines **50** are arranged at intervals in the second direction oy. The conductive material may be, for example, a low-resistance metal material or metal oxide, such as W or a compound thereof.

(37) Further referring to FIG. **13** and FIG. **14**, a capacitor (not shown in the figure) may further be formed on the surface of the VGAA transistor **40**. Specifically, the capacitor is located on the top surface of the body pillar **12** and is connected to one of the source of the transistor **40** or the drain of the transistor **40**, and the other of the source of the transistor **40** or the drain of the transistor **40** is connected to the bit line **30**. After that, the remaining gaps may be filled with a second isolation layer **60**, to electrically isolate the devices formed on the substrate **100** from each other. The second

isolation layer **60** may be made of, for example, silicon oxide, silicon nitride, or silicon oxynitride.

(38) In the embodiments of the present disclosure, a plurality of body structures **10** are formed on a substrate, wherein each body structure **10** includes a body layer **11** and a plurality of discrete body pillars **12** located on the body layer **11**; the part of the body layer **11** located between adjacent body pillars **12** is etched to form a plurality of openings **11a** in the body layer **11**, so as to form a continuous metal silicide in the body layer **11** to serve as a bit line **30**, thereby reducing the resistance of the semiconductor structure and improving the performance of the semiconductor structure. In addition, in the embodiments of the present disclosure, the body layer **11** are further doped with ions through the plurality of openings **11a**, to reduce contact resistance between silicon in the body pillars **12** or body layer **11** and the metal silicide in the process of forming the metal silicide in the body layer **11**. The embodiments of the present disclosure further illustrate a semiconductor structure in which VGAA transistors **40**, word lines **50**, capacitors and an isolation structure are manufactured after the buried bit lines **30** are formed. The semiconductor structure provided by the embodiments of the present disclosure effectively reduces the size of the memory and improves the level of integration and performance of the memory.

(39) As shown in FIG. **15** to FIG. **23**, in other embodiments of the present disclosure, to adapt to the versatility and flexibility of the manufacturing process, it is also possible to form the bit lines **30** after forming of the VGAA transistors **40** and the word lines **50**.

(40) It should be noted that, FIG. **16** and FIG. **23** are partial cross-sectional views taken along a first direction *ox*; and FIG. **17** to FIG. **22** are partial cross-sectional views taken along a second direction *oy*.

(41) Referring to FIG. **15**, an embodiment of the present disclosure provides a method of manufacturing a semiconductor structure, including:

(42) **S20**: Provide a substrate **100**.

(43) **S21**: Etch the substrate **100** to form a plurality of body structures **10** arranged at intervals in a first direction *ox*, wherein a space between adjacent body structures **10** is filled with a first isolation layer **20**, each body structure **10** includes a body layer **11** and a plurality of discrete body pillars **12** located on the body layer **11**, the plurality of body pillars **12** are arranged at intervals along a second direction *oy*, and the first direction *ox* is different from the second direction *oy*.

(44) **S22**: Form gates **41** of a plurality of VGAA transistors **40** and a plurality of word lines **50** on sidewalls of the plurality of body pillars **12** by using a self-aligned process, wherein space between adjacent word lines **50** is filled with a third isolation layer **70**.

(45) **S23**: Etch back part of the third isolation layer **70** to expose the body pillars **12**.

(46) **S24**: Form a protective layer **13** on sidewalls of the exposed body pillars **12**, wherein gaps **13a** exist between parts of the protective layer **13** of adjacent body pillars.

(47) **S25**: Remove a remaining part of the third isolation layer **70** along the gaps **13a** until the body layer **11** is exposed.

(48) **S26**: Etch an exposed part of the body layer **11** along the gaps **13a**, to form a plurality of openings **11a** in the body layer **11**.

(49) **S27**: Siliconize the body layer **11** through the plurality of openings **11a** to form a metal silicide that is continuous along the second direction *oy*, so as to form a plurality of bit lines **30**, each bit line **30** is in the corresponding body layer **11**.

(50) Specifically, referring to FIG. **15** and FIG. **2** to FIG. **8**, in some embodiments of the present disclosure, steps **S20** and **S21** may be the same as steps **S10** and **S11**, and are not described in detail again herein.

(51) Further referring to FIG. **16** and FIG. **17**, in step **S22**, gates **41** of the VGAA transistors **40** and the word lines **50** that connect the gates **41** are formed on the sidewalls of the plurality of body pillars **12** by using a self-aligned process. The gate **41** surrounds the body pillar **12**. The body pillar **12** is formed into a source, a drain and a vertical channel of the VGAA transistor **40**. The gate **41** surrounds the vertical channel, and includes a part of a gate dielectric layer **411** and a part of a gate

conductive layer **412** that surrounds the channel in sequence from the inner side to the outer side. Each word line **50** connects the gates **41** and extends along the first direction ox. The word lines **50** are arranged at intervals along the second direction oy. Space between adjacent word lines **50** is filled with a third isolation layer **70**. Parts of the third isolation layer **70** extend along the first direction ox and are arranged at intervals along the second direction oy. In some embodiments, the third isolation layer **70** is made of, for example, silicon nitride. The VGAA transistor **40** provided by the embodiments of the present disclosure has a small size. More transistor devices can be formed per unit area, to improve the level of integration of the semiconductor structure, such that the semiconductor structure is adapted to the miniaturization, light weight, and energy saving requirements of end customers.

(52) Referring to FIG. **18**, in step S23, the third isolation layer **70** is partially removed to reach a top surface of the gate **41**, so as to expose a part of the body pillar **12** at the source or drain in the upper part of the VGAA transistor **40**, and a top surface of the remaining part of the third isolation layer **70** is lower than the top surface of the gate **41**.

(53) Referring to FIG. **19**, in step S24, the protective layer **13** is formed on the sidewalls of the exposed body pillars **12**. In some embodiments, for example, after an oxide is deposited on the surfaces of the exposed body pillars **12** to project the silicon sidewalls of the sources/drains on the body pillars **12**, the oxide on the top surfaces of the body pillars **12** and the top surface of the third isolation layer **70** may be removed through dry etching. In this case, the surfaces of the body pillars **12** are protected by the protective layer **13** and a fourth isolation layer **80** that is reserved on the body pillars **12** in the self-aligned process. The gaps **13a** exist between parts of the protective layer **13** of adjacent body pillars **12**. For example, the material of the protective layer **13** may be same as the material of the fourth isolation layer **80**, and may be, for example, silicon oxide.

(54) Further referring to FIG. **19**, in step S24, the remaining part of the third isolation layer **70** is removed along the gaps **13a**, until the body layer **11** is exposed. In some embodiments of the present disclosure, in order to remove the remaining part of the isolation layer **70** completely, the material of the remaining part of the isolation layer **70** may be removed along the gaps **13a** by using a wet etching process, to expose the part of the body layer **11** between adjacent body pillars **12**. The remaining part of the isolation layer **70** may be partially removed by using a wet etching process. The concentration, flow or etching time of an etching solution may be adjusted. The etching solution, for example, includes but is not limited to, a single solution of or a mixture of more than one of the following solutions: phosphoric acid, diluted hydrofluoric acid, diluted sulfuric acid, carbon tetrafluoride (CF₄), or sulfur hexafluoride (SF₆).

(55) Referring to FIG. **20**, in step S25, the exposed part of the body layer **11** is etched along the gaps **13a**, to form a plurality of openings **11a** in the body layer **11**. In some embodiments, the process of etching the exposed part of the body layer **11** may be the forming process described above, but is not limited thereto. Definitely, the etching process may be implemented in any possible manner. Any method that can form the expected openings **11a** in the body layer **11** shall fall within the protection scope of the present disclosure.

(56) Further referring to FIG. **22** and FIG. **23**, step S26 is then performed: a conductive material is deposited in the openings **11a** along the gaps **13a**, and an annealing process is performed, such that the conductive material reacts with the silicon material of the body layer **11**, so as to form a metal silicide that is continuous in the body layer **11** to serve as the bit line **30**. Further, in some embodiments, referring to FIG. **21**, before the bit line **30** is formed, in order to reduce the Schottky resistance at the interface between the metal silicide and silicon, low-energy high-dosage ion implant may be performed on the body layer **11** along the gaps **13a** by using the openings **11a**. For example, ion implant may be implemented through ion implantation using P-type ions, such as B ions. Definitely, in other embodiments, N-type ions that have a higher current may be used. Specifically, for example, As ions or P ions may be used. The preset energy for ion implantation may be, for example, 1 keV to 20 keV; and/or the preset dosage is 1e^{sup.15} cm^{sup.-2} to

1e.sup.18 cm.sup.-2. For example, the preset energy is 1 keV, 1.5 keV, 5 keV, 10 keV or 12 keV; the preset dosage is 1e.sup.15 cm.sup.-2, 5e.sup.15 cm.sup.-2, 1e.sup.16 cm.sup.-2, 1e.sup.17 cm.sup.-2 or 1e.sup.18 cm.sup.-2. Further, during the ion implantation process, the source/drain of the body pillar **12** may be doped at the same time, to reduce steps of the manufacturing process. Definitely, doping may alternatively be performed before or after the ion implantation process, which may be adjusted according to actual needs.

(57) In this case, in the embodiments of the present disclosure, the VGAA transistors **40** and the word lines **50** are formed before the bit lines **30** are formed, which is adaptable to the versatility of the manufacturing process. Then, the protective layer **13**, the fourth isolation layer **80** and the like in the foregoing process can further be removed, and a capacitor (not shown in the figure) may be formed on the surface of the VGAA transistor **40**. Specifically, the capacitor is located on the top surface of the body pillar **12** and is connected to one of the source of the transistor **40** or the drain of the transistor **40**, and the other of the source of the transistor **40** or the drain of the transistor **40** is connected to the bit line **30**. After that, the remaining gaps are filled with a second isolation layer **60**, to electrically isolate the devices formed on the substrate **100** from each other.

(58) Referring to FIG. **1** to FIG. **23**, an embodiment of the present disclosure provides a semiconductor structure, including a substrate **100**, body structures **10**, a first isolation layer **20**, and bit lines **30**. The body structures **10** are arranged at intervals along a first direction ox, and space between adjacent body structures **10** is filled with the first isolation layer **20**. Each body structure **10** includes a body layer **11** and a plurality of discrete body pillars **12** located on the body layer **11**. The plurality of body pillars **12** are arranged at intervals along a second direction oy, wherein a part of the body layer **11** between adjacent body pillars **12** is provided with openings **11a**. The bit lines **30** each include a metal silicide that is continuous along the second direction oy in a corresponding body layer **11** through the openings **11a**.

(59) In further embodiments of the present disclosure, the semiconductor structure further includes vertical transistors **40**, word lines **50**, capacitors (not shown in the figure), and an isolation structure that includes the first isolation layer **20** and the second isolation layer **60**. The gate of the VGAA transistor **40** surrounds the sidewall of the body pillar **12**, and the body pillar **12** is formed into a drain, a vertical channel, and a source of the VGAA transistor **40**. The gate of the VGAA transistor **40** is connected to the word line **50**. The word lines **50** extend along the first direction ox and are arranged at intervals in the second direction oy. One of the source of the VGAA transistor **40** or the drain of the VGAA transistor **40** is connected to the bit line **30**, and the other is connected to the capacitor.

(60) The semiconductor structure provided by the embodiments of the present disclosure may be formed by using the manufacturing method provided above, but is not limited thereto. In the semiconductor structure provided by the foregoing embodiment, a plurality of body structures are formed on a substrate, wherein each body structure includes a body layer and a plurality of discrete body pillars located on the body layer; the part of the body layer located between adjacent body pillars is etched to form a plurality of openings in the body layer, so as to facilitate lateral growth of the metal silicide in the substrate, such that the metal silicide at adjacent positions becomes continuous to form a line. In this way, a continuous metal silicide is formed in the body layer to serve as a bit line, thereby reducing the resistance of the semiconductor structure and improving the performance of the semiconductor structure. In addition, in the embodiments of the present disclosure, ions are implanted into the body layer through the plurality of openings, to reduce contact resistance between silicon and the metal silicide in the process of forming the metal silicide in the body layer.

(61) It may be noted that the foregoing embodiments are merely for the purpose of description instead of limiting the present disclosure.

(62) The embodiments of this specification are described in a progressive manner, and each embodiment focuses on differences from other embodiments. The same or similar parts between

the embodiments may refer to each other.

(63) The technical characteristics of the above embodiments can be employed in arbitrary combinations. To provide a concise description of these embodiments, all possible combinations of all the technical characteristics of the above embodiments may not be described; however, these combinations of the technical characteristics should be construed as falling within the scope defined by the specification as long as no contradiction occurs.

(64) The above embodiments are only intended to illustrate several implementations of the present disclosure in detail, and they should not be construed as a limitation to the patentable scope of the present disclosure. It should be noted that those of ordinary skill in the art can further make variations and improvements without departing from the conception of the present disclosure. These variations and improvements all fall within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure should be subject to the protection scope defined by the claims.

Claims

1. A method of manufacturing a semiconductor structure, comprising: providing a substrate; etching the substrate to form a plurality of body structures arranged at intervals in a first direction, wherein a space between adjacent body structures of the plurality of body structures is filled with a first isolation layer, each body structure of the plurality of body structures comprises a respective body layer and a respective plurality of body pillars that are discrete and are located on the respective body layer, the respective plurality of body pillars are arranged at intervals along a second direction, and the first direction is different from the second direction; in each body structure of the plurality of body structures, etching a part of the respective body layer between adjacent body pillars of the respective plurality of body pillars to form a plurality of discrete openings in the respective body layer; and in each body structure of the plurality of body structures, siliconizing the respective body layer through the respective plurality of discrete openings to form a metal silicide that is continuous in the second direction along the respective plurality of body pillars, so as to form a bit line being in the respective body layer.
2. The method of manufacturing according to claim 1, wherein each discrete opening of the plurality of discrete openings is an arc-shaped opening, and partially overlaps with two respective body pillars of the plurality of body pillars that are adjacent to the respective opening.
3. The method of manufacturing according to claim 2, wherein a maximum opening distance of the arc-shaped opening in the body layer is 8.5 nm to 11.5 nm.
4. The method of manufacturing according to claim 2, wherein the etching a part of the respective body layer between adjacent body pillars of the respective plurality of body pillars to form a plurality of discrete openings in the respective body layer comprises: forming a protective layer on sidewalls of the respective plurality of body pillars, wherein gaps exist between parts of the protective layer of the adjacent ones of the body pillars of the respective plurality of body pillars; and performing an isotropic etching on the respective body layer along the gaps to form the plurality of discrete openings in the respective body layer.
5. The method of manufacturing according to claim 4, wherein the isotropic etching is performed by using a reactive etching gas.
6. The method of manufacturing according to claim 1, wherein the siliconizing the respective body layer through the respective plurality of discrete openings to form a metal silicide that is continuous along the second direction comprises: depositing a conductive material on surfaces of the respective plurality of openings, and performing an annealing process to form the bit line.
7. The method of manufacturing according to claim 6, before the siliconizing the respective body layer through the respective plurality of discrete openings to form a metal silicide that is continuous along the second direction, the method further comprises: performing an ion implant on the

respective body layer along the respective plurality of discrete openings with a preset energy and a preset dosage to form a doped region, and performing an annealing process; and siliconizing the respective body layer having the doped region.

8. The method of manufacturing according to claim 7, wherein at least one of: the preset energy is 1 keV to 20 keV; or the preset dosage is $1 \times 10^{15} \text{ cm}^{-2}$ to $1 \times 10^{18} \text{ cm}^{-2}$.

9. The method of manufacturing according to claim 1, wherein the etching the substrate to form a plurality of body structures comprises: etching the substrate along the first direction and forming a plurality of first trenches, and filling the first trenches with the first isolation layer; and etching the substrate and the first isolation layer along the second direction and forming a plurality of second trenches, wherein an etching depth of the second trench is less than an etching depth of the first trench to form the plurality of body structures arranged at intervals along the first direction on the substrate, wherein each body structure of the plurality of body structures comprises the respective body layer and the respective plurality of body pillars that are discrete and are located on the respective body layer, and the respective plurality of body pillars are arranged at the intervals along the second direction and spaced apart from each other by the plurality of second trenches.

10. The method of manufacturing according to claim 1, further comprising: in each body structure of the plurality of body structures, forming a plurality of vertical gate-all-around transistors, wherein a gate of each vertical gate-all-around transistor of the plurality of vertical gate-all-around transistor surrounds a body pillar of the respective plurality of body pillars; forming a plurality of word lines, wherein each word line of the plurality of word lines connects the gates of the plurality of vertical gate-all-around transistors of adjacent body structures of the plurality of body structures; forming a plurality of capacitors, wherein each capacitor of the plurality of capacitors is connected to a source of a respective vertical gate-all-around transistor of the plurality of vertical gate-all-around transistors or a drain of the respective vertical gate-all-around transistor of the plurality of vertical gate-all-around transistors; and filling gaps with a second isolation layer.

11. The method of manufacturing according to claim 10, wherein the source of the respective vertical gate-all-around transistor of the plurality of vertical gate-all-around transistors, the drain of the respective vertical gate-all-around transistor of the plurality of vertical gate-all-around transistors, and vertical channel of the respective vertical gate-all-around transistor of the plurality of vertical gate-all-around transistors are sequentially arranged by doping a respective body pillar of the respective plurality of body pillars with ions of a same type.

12. The method of manufacturing according to claim 1, after the etching the substrate to form a plurality of body structures, the method further comprises: forming gates of a plurality of vertical gate-all-around transistors and a plurality of word lines on sidewalls of body pillars of the plurality of body pillars by using a self-aligned process, wherein a space between adjacent word lines of the plurality of word lines is filled with a third isolation layer; etching back a part of the third isolation layer to expose portions of the body pillars of the plurality of body pillars; forming a protective layer on sidewalls of the portions of the body pillars that are exposed, wherein gaps exist between parts of the protective layer of adjacent body pillars of the plurality of body pillars; removing a remaining part of the third isolation layer along the gaps until the plurality of body layers is exposed; etching an exposed part of the plurality of body layers along the gaps, to form the plurality of discrete openings in each of the body structures of the plurality of body structures.

13. The method of manufacturing according to claim 12, further comprising: forming a plurality of capacitors, wherein each capacitor of the plurality of capacitors is connected to a source of a vertical gate-all-around transistor of the plurality of vertical gate-all-around transistor or a drain of the vertical gate-all-around transistor of the plurality of the vertical gate-all around transistor; and filling gaps with a second isolation layer.
